# ADVERSE DRUG REACTION (ADR) DATA ANALYSIS REPORT

Frequency profiling • Severity screening • Age distribution • Safety signal summary

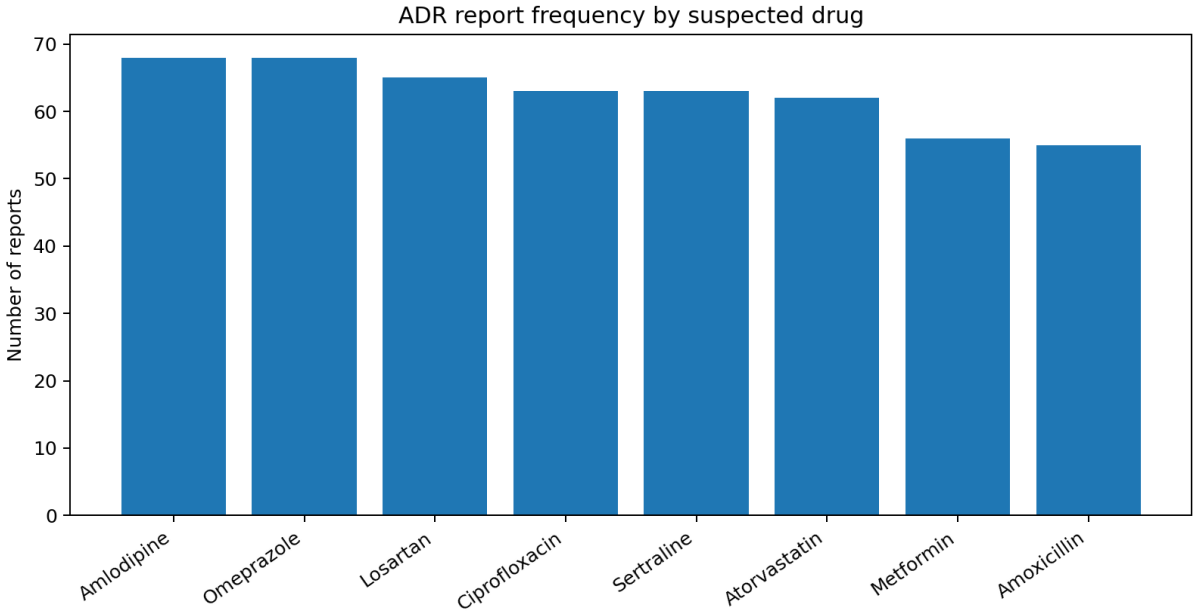
## 1. Dataset overview

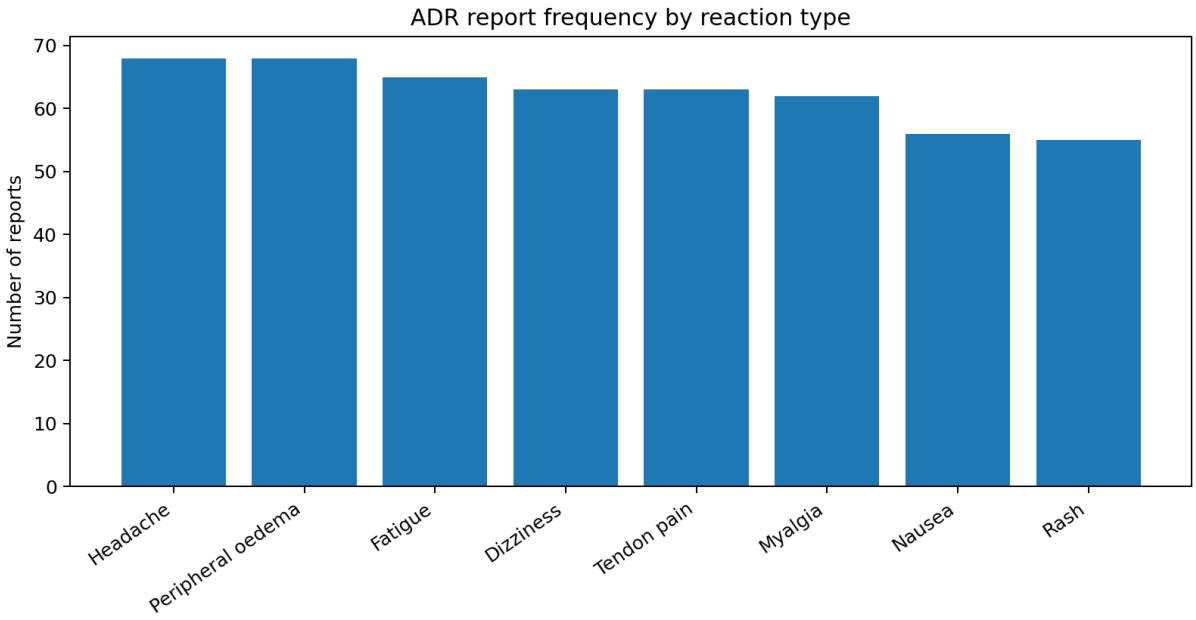
| Metric | Result |
|---|---|
| Source file | adverse_drug_reactions_v2(1).csv |
| Total reports | 500 |
| Variables | 19 original columns (plus derived age bracket) |
| Reporting period | 01 Jan 2026 to 28 Aug 2026 |
| Missing values | 0 across the supplied columns |
| Serious reports | 81 (16.2%) |
| Unexpected reports | 65 (13.0%) |

The supplied file contains 500 records and 18 fields. The supplied report IDs use the SYN-ADR prefix, so results should be treated as an analysis of this supplied dataset rather than verified spontaneous-reporting data.

## 2. Reporting frequency index

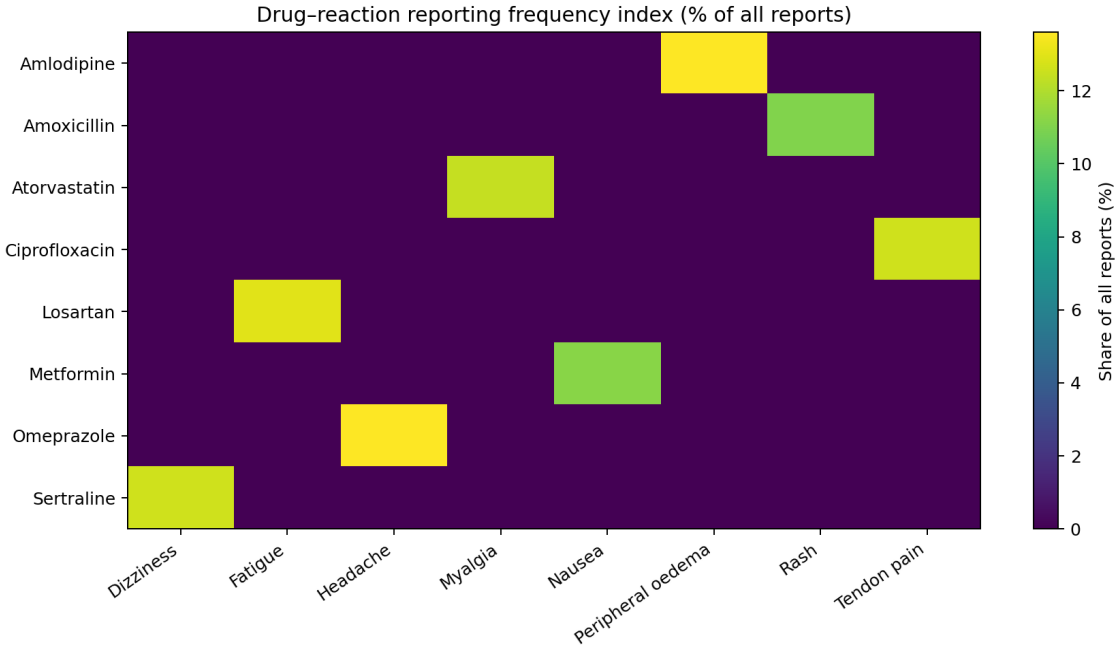
For each drug or reaction type, reporting frequency (%) = number of reports for that category ÷ all reports × 100. The drug–reaction matrix reports each pair as a percentage of all records.






Most frequently reported suspected drugs: Amlodipine, Omeprazole, Losartan. Highest individual drug frequency: Amlodipine (68 reports; 13.6%). Most frequently reported reactions: Headache, Peripheral oedema, Fatigue.

## 3. Drug–reaction reporting frequency index




The matrix shows one observed drug–reaction pairing for each of the eight drugs in this dataset. No alternative reaction pairings are represented in the supplied records. This limits comparative signal detection across different drug–reaction combinations.

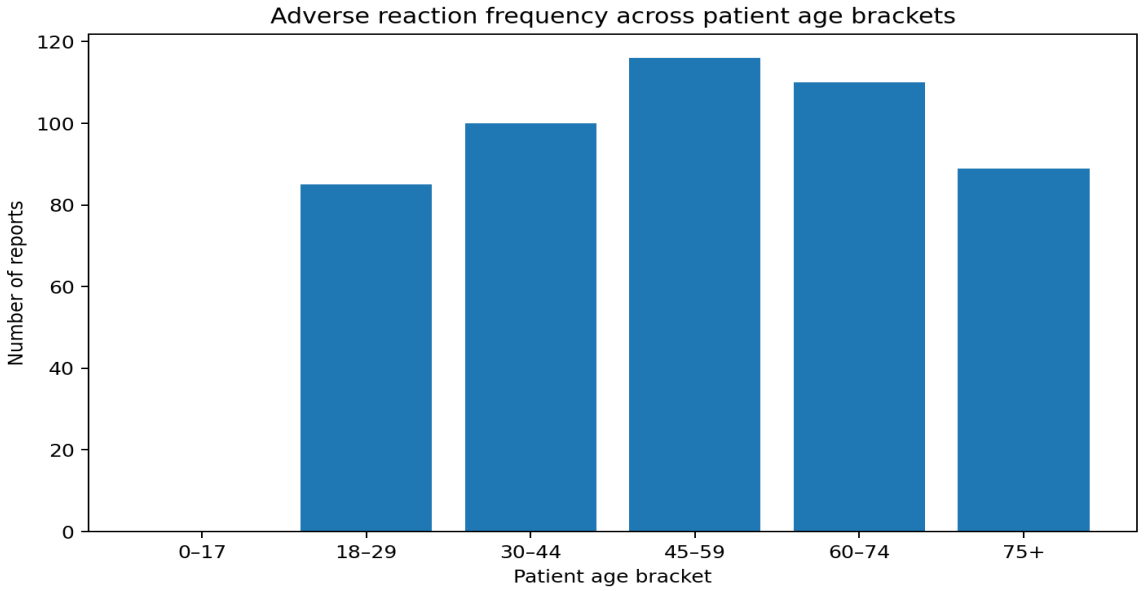
## 4. High-risk combination screening

Severity ratio (%) = serious reports for a drug–reaction pair ÷ all reports for that pair × 100. To avoid ranking isolated one-off pairs, the screening table includes pairs with at least three reports. This is a descriptive prioritization rule, not a validated regulatory threshold.

| Suspected drug | Reaction | Reports | Serious | Severity ratio |
|---|---|---|---|---|
| Omeprazole | Headache | 68 | 15 | 22.1% |
| Amlodipine | Peripheral oedema | 68 | 14 | 20.6% |
| Atorvastatin | Myalgia | 62 | 11 | 17.7% |
| Sertraline | Dizziness | 63 | 11 | 17.5% |
| Losartan | Fatigue | 65 | 11 | 16.9% |
| Ciprofloxacin | Tendon pain | 63 | 9 | 14.3% |
| Metformin | Nausea | 56 | 7 | 12.5% |
| Amoxicillin | Rash | 55 | 3 | 5.5% |

Within this dataset, Omeprazole–Headache has the highest observed severity ratio (22.1%; 15/68 reports). Amlodipine–peripheral oedema is next (20.6%). These are candidates for review, not proof of causality or comparative drug risk.

## 5. ADR frequency across patient age brackets




The 45–59 year bracket contains the most reports (116; 23.2%). No records were present for patients aged 0–17. These are report counts, not age-specific incidence rates; population denominators and exposure data are unavailable.

## 6. Safety signal detection summary

**Overall profile:** 500 reports were analysed; 81 (16.2%) were marked serious and 65 (13.0%) were marked unexpected. Among serious reports, 35 listed hospitalisation and 46 listed another medically important condition.

**Priority review candidates:** Omeprazole–headache (22.1% serious), Amlodipine–peripheral oedema (20.6%), Atorvastatin–myalgia (17.7%), and Sertraline–dizziness (17.5%) had the highest observed severity ratios among pairs with at least three reports.

**Interpretation:** The dataset has a one-to-one mapping between each suspected drug and a single reaction type. Therefore, it does not support robust comparison of alternative reactions within the same drug, nor conventional disproportionality methods such as reporting odds ratio (ROR) or proportional reporting ratio (PRR), which require suitable comparator counts. No exposure denominators are supplied, so incidence or absolute risk cannot be estimated. Seriousness ratio is not a causal risk estimate.

**Recommended follow-up:** Validate the source and coding; clinically review serious and unexpected cases; examine case narratives, time-to-onset, dechallenge/rechallenge, concomitant medicines, dose, indication and outcomes; compare against a broader reference database with appropriate controls; and assess signals using validated pharmacovigilance methods before drawing conclusions.

## 7. Methodology, assumptions and limitations

• All original columns were checked for missing values; none were found in this file.

• Age bands used: 0–17, 18–29, 30–44, 45–59, 60–74 and 75+ years.

• Seriousness ratio uses the supplied Serious field (Yes/No).

• High-risk screening displays pairs with at least three reports; the threshold is exploratory and selected for readability.

• Counts and percentages describe this dataset only. They must not be interpreted as population incidence, comparative safety, or causal association.

• The report does not replace medical review, signal validation, or regulatory assessment.

Prepared from the user-provided CSV dataset. Analysis is descriptive and exploratory.